

Current

Building Models to Describe our World
Electric Current, Ohm's Law, and Power



Outline

Electric Current

- Microscopic model of current

- Drift velocity

- Current density

 - At the microscopic level

 - From drift velocity

Ohm's Law

- Macroscopically

- Resistance

- Combining resistors

- Resistivity

 - Resistivity versus temperature

 - Resistivity for a few materials

- Measuring resistance

- Resistance versus temperature

Electrical Power

- Batteries

- The electric cell

- Power dissipated in a resistor

- Electric power

- Household appliances

- AC voltage and current

- Power dissipated in a resistor with AC current

- RMS current

 - RMS voltage

Electric Current

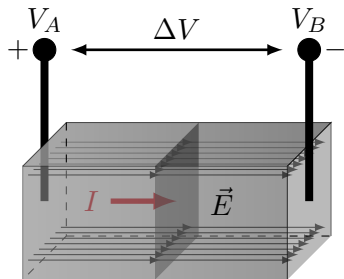
Building Models to Describe our World
Electric Current, Ohm's Law, and Power

Electric current

- Electric current is a measure of the rate at which charge flows through a material:

$$I = \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

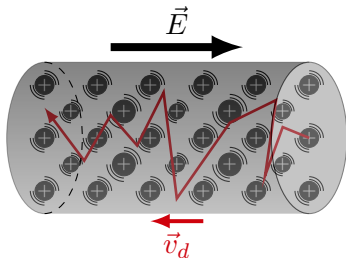
- Current is positive in the direction in which positive charges flow. In reality, it is **almost always electrons that flow**, so current is usually in the *opposite* direction of the flow of the actual charges.
- Electric current is a *macroscopic* quantity.



We apply a potential difference, ΔV , across a metal and current, I , flows.

Microscopic model of current

- We can model current in a metal using a *microscopic* model of charges “drifting” through the metal with an average velocity:
 - Typical **drift velocity**: $v_d \sim 1 \times 10^{-4}$ m/s
- In reality, electrons are moving around much faster, in mostly random directions and bouncing off of atoms:
 - Typical speed of electrons: 1×10^8 m/s

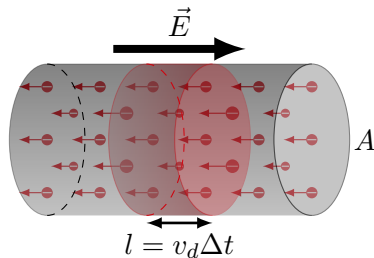


An electron moving through a resistor and colliding with atoms, resulting in heat. The average speed of the electron is constant as the work done by the electric field is converted into heat.

Current and drift velocity

We have, n , electrons per unit volume, with charge q , moving with average speed, v_d , through a resistor of cross-section, A :

- In time Δt , electrons move a distance $l = v_d \Delta t$.

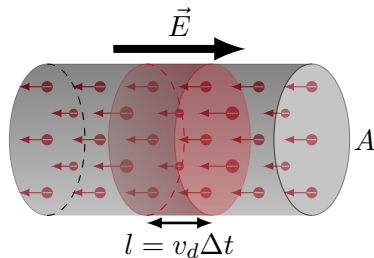


Electrons drifting with an average velocity, $\vec{v}_d$

Current and drift velocity

We have, n , electrons per unit volume, with charge q , moving with average speed, v_d , through a resistor of cross-section, A :

- In time Δt , electrons move a distance $l = v_d \Delta t$.
- This corresponds to a volume of electrons, $V = lA = v_d \Delta t A$

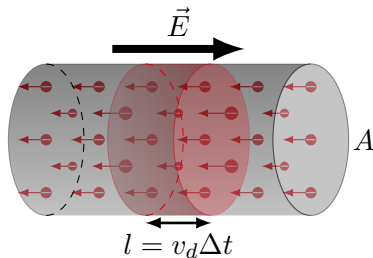


Electrons drifting with an average velocity, $\vec{v}_d$

Current and drift velocity

We have, n , electrons per unit volume, with charge q , moving with average speed, v_d , through a resistor of cross-section, A :

- In time Δt , electrons move a distance $l = v_d \Delta t$.
- This corresponds to a volume of electrons, $V = lA = v_d \Delta t A$
- with a total charge, $\Delta Q = Vnq = v_d \Delta t Anq$

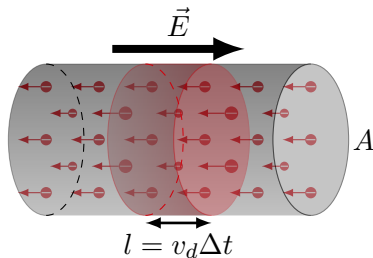


Electrons drifting with an average velocity, $\vec{v}_d$

Current and drift velocity

We have, n , electrons per unit volume, with charge q , moving with average speed, v_d , through a resistor of cross-section, A :

- In time Δt , electrons move a distance $l = v_d \Delta t$.
- This corresponds to a volume of electrons, $V = lA = v_d \Delta t A$
- with a total charge, $\Delta Q = Vnq = v_d \Delta t Anq$
- resulting in a current, I :
$$I = \frac{\Delta Q}{\Delta t} = v_d Anq$$



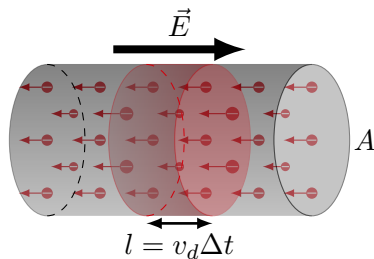
Electrons drifting with an average velocity, $\vec{v}_d$

Current and drift velocity

We have, n , electrons per unit volume, with charge q , moving with average speed, v_d , through a resistor of cross-section, A :

- In time Δt , electrons move a distance $l = v_d \Delta t$.
- This corresponds to a volume of electrons, $V = lA = v_d \Delta t A$
- with a total charge, $\Delta Q = Vnq = v_d \Delta t Anq$
- resulting in a current, I :
$$I = \frac{\Delta Q}{\Delta t} = v_d Anq$$

This allows us to relate a macroscopic quantity (I , current) to microscopic quantities (e.g. v_d).



Electrons drifting with an average velocity, $\vec{v}_d$

Current and current density

- We can model charges moving through a metal macroscopically, by using current, I :

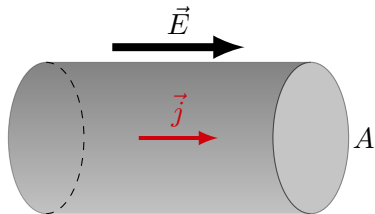
$$I = \frac{dQ}{dt}$$

This is the amount of charge flowing through cross section, A , of the metal per unit time.

- We can define the current density, as the current per unit area, j :

$$j = \frac{I}{A}$$

j , the current density can depend on position and is a microscopic quantity.



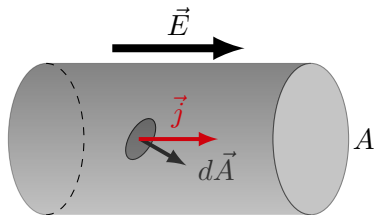
Current density is a vector parallel to the electric field (and anti-parallel to drift velocity for electrons).

Current density at the microscopic level

- We define the **current density, $\vec{j}$, as a vector parallel to the electric field**, corresponding to the direction of the average flow of positive charges.
- If we choose a surface, the current through that surface is that part of the current density flowing perpendicular to the surface:

$$I = \int \vec{j} \cdot d\vec{A} = \int j dA_{\perp}$$

(just like a “flux”; in fact, exactly a flux!)



Current density is a vector parallel to the electric field (and anti-parallel to drift velocity for electrons).

Current density from drift velocity

- The current:

$$\vec{I} = -enA\vec{v}_d$$

- The current density:

$$\vec{j} = \frac{\vec{I}}{A} = -en\vec{v}_d$$

Here, we have explicitly chosen negative charges (electrons with charge $q = -e = -1.6 \times 10^{-19}$ C). Current and current density are in the opposite direction of the drift velocity because q is negative.

Ohm's Law

Building Models to Describe our World
Electric Current, Ohm's Law, and Power

Ohm's Law

- For most materials:

$$\vec{j} = \sigma \vec{E}$$

(We call these “ohmic” materials, because the law holds).

- σ is the **conductivity** of the material; it is a material-dependent measure of how large a current density one gets for a given applied electric field.

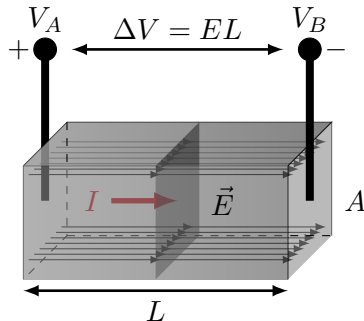
Ohm's Law, macroscopically

- Microscopically:

$$\vec{j} = \frac{\vec{I}}{A} = -en\vec{v}_d \quad \vec{j} = \sigma \vec{E}$$

- Relation between current and current density, on a *macroscopic* piece of metal:

$$\frac{I}{A} = \sigma E = \sigma \frac{\Delta V}{L}$$
$$\therefore \Delta V = \frac{1}{\sigma} \frac{L}{A} I$$



We apply a potential difference, ΔV , across a metal and current, I , flows due to the electric field, $\vec{E}$.

Ohm's Law, macroscopically

- Microscopically:

$$\vec{j} = \frac{\vec{I}}{A} = -en\vec{v}_d \quad \vec{j} = \sigma \vec{E}$$

- Relation between current and current density, on a *macroscopic* piece of metal:

$$\begin{aligned} \frac{I}{A} &= \sigma E = \sigma \frac{\Delta V}{L} \\ \therefore \Delta V &= \frac{1}{\sigma} \frac{L}{A} I \\ \boxed{\Delta V} &= \color{red}{R} I \end{aligned}$$

Resistance

We call, R , the “resistance” of the piece of material (a “resistor”).

$$R = \frac{1}{\sigma} \frac{L}{A} = \rho \frac{L}{A}$$

where ρ is the “resistivity” and σ the “conductivity” of the material.

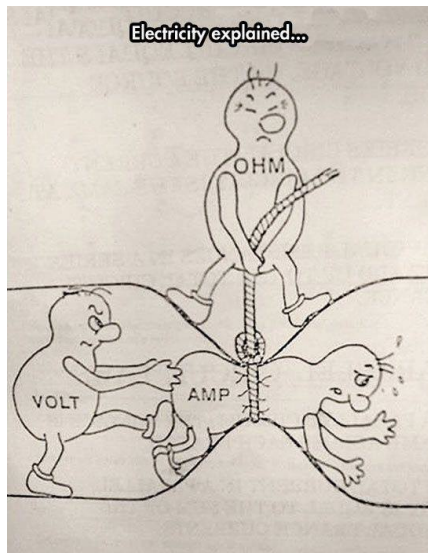
Resistance is the macroscopic property of resistor of length, L , area, A , made from a material with resistivity, ρ .

Ohm's Law

- Technically, Ohm's Law refers to the microscopic version, $\vec{j} = \sigma \vec{E}$.
- Ohm noted that the current through a material is proportional to the electric potential difference, introducing “resistance”, R , as the constant of proportionality:

$$\Delta V \propto I = RI$$

- For any “resistor” can use Ohm's Law to find the current through it, given a potential difference.
- High resistance = low current for a given electric field



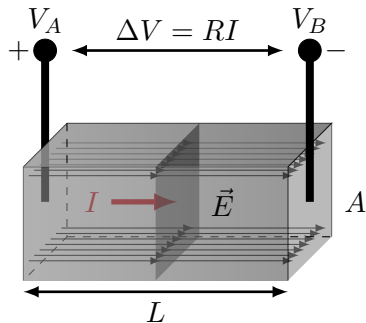
Resistance

- Ohm's law for a resistor:

$$\Delta V = RI$$

- The resistance depends on the shape, (L, A) , and the material, ρ :

$$R = \rho \frac{L}{A}$$



We apply a potential difference, ΔV , across a resistor, R , and current, I , flows.

Resistance

- Ohm's law for a resistor:

$$\Delta V = RI$$

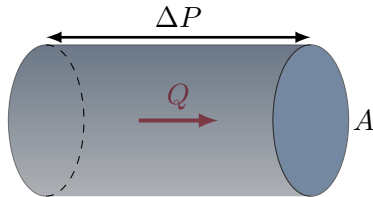
- The resistance depends on the shape, (L, A) , and the material, ρ :

$$R = \rho \frac{L}{A}$$

- Recall Poiseuille flow (fluid with viscosity):

$$\Delta P = RQ$$

- For a horizontal pipe: $R = \frac{8\eta L}{\pi r^4}$



A fluid flowing through a pipe due to a pressure difference.

Analogy

The math for current flowing through a resistor is the same as for fluid flowing through a pipe! Voltage is like pressure difference.

Combining resistors

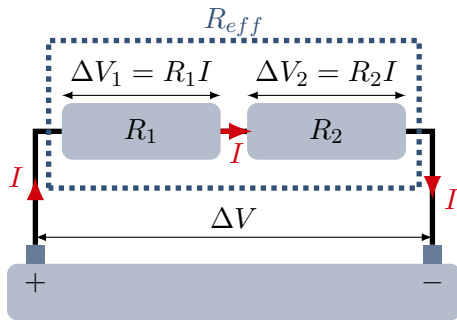
In series: The voltages across each resistor add up to the battery voltage:

$$\Delta V_1 + \Delta V_2 = \Delta V$$

$$R_1 I + R_2 I = R_{eff} I$$

Like making a longer resistor, total resistance increases:

$$R_{eff} = R_1 + R_2$$



R₁ and R₂ in series.

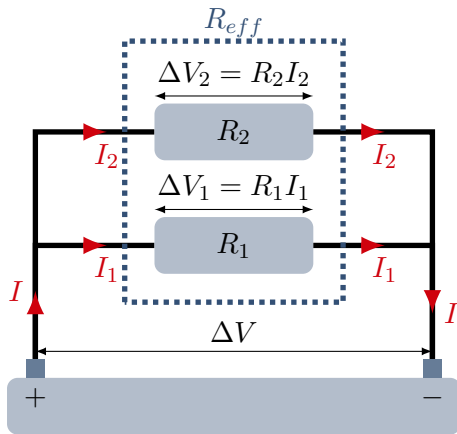
Combining resistors

In parallel: The currents through each resistor add up to the current through the battery:

$$I_1 + I_2 = I$$
$$\frac{\Delta V}{R_1} + \frac{\Delta V}{R_1} = \frac{\Delta V}{R_{eff}}$$

Like making a wider resistor, total resistance decreases:

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}$$



R₁ and R₂ in parallel.

Resistivity

- Resistivity is a temperature-dependent property of a material:

$$R = \rho(T) \frac{L}{A}$$

- The temperature dependence, is “often” linear (for most materials, for common ranges of temperatures):

$$\rho(T) = \rho_0 [1 + \alpha(T - T_0)]$$

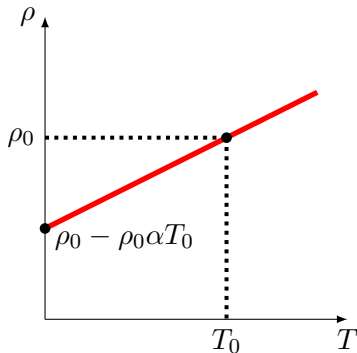
- We use a reference resistivity, ρ_0 , at a reference temperature, T_0 , (usually $T_0 = 20^\circ\text{C}$).
- Note that α , the temperature coefficient, depends on the choice of reference temperature, T_0 .

Resistivity versus temperature

- The resistivity is linear in temperature:

$$\rho(T) = \rho_0 [1 + \alpha(T - T_0)]$$

- Remember, conductors resist the flow of current, because there are atoms in the conductor that the charges run into.
- If those atoms are shaking around more, it's more likely for the flowing charges to run into them.
 - More shaking $\rightarrow$ higher temperature.
 - More running into atoms $\rightarrow$ higher resistance.



Resistivity as a function of temperature. It's a silly parametrization, because the slope is $\rho_0 \alpha$ instead of just α .

Resistivity for a few materials

Material	Resistivity, ρ [$\Omega \cdot \text{m}$]	Temperature coefficient, α [$^{\circ}\text{C}^{-1}$]	Free electron density, n [m^{-3}]
Silver	1.59×10^{-8}	0.0038	5.86×10^{28}
Copper	1.68×10^{-8}	0.0040	8.46×10^{28}
Gold	2.44×10^{-8}	0.0034	5.90×10^{28}
Aluminum	2.74×10^{-8}	0.0039	18.1×10^{28}
Iron	9.70×10^{-8}	0.0050	17.0×10^{28}
Silicon	0.1-1000	-0.0750	0
Rubber	$(1-100) \times 10^{13}$	0	0
Quartz	7.5×10^{17}	0	0

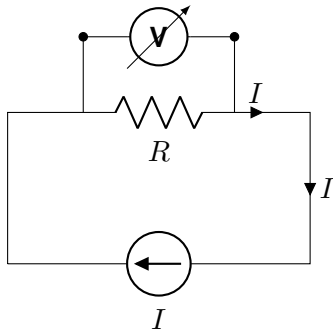
Table: Resistivity, free electron density and temperature coefficients of common materials. All properties are listed for a reference temperature of 20°C .

Measuring resistance

- Instead of a battery, we use a constant current source
- We can measure the voltage across the resistor to find its resistance:

$$\Delta V = RI$$

- What is the resistance at room temperature ?

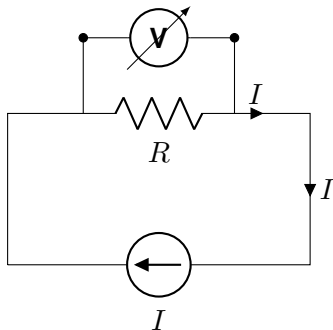


Constant current source connected to a resistor. A voltmeter measures the voltage, ΔV , across the resistor.

Resistance versus temperature

What is the resistance

- At 20°C (room temperature)?
- At 0°C (ice water)?
- At 100°C (boiling water)?
- At -196°C (liquid nitrogen)?



Constant current source connected to a resistor. A voltmeter measures the voltage, ΔV , across the resistor.

Resistance versus temperature

What is the resistance

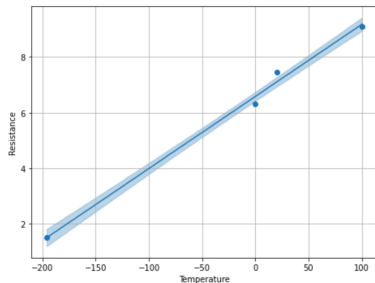
- At -196°C (liquid nitrogen)? $1.32\ \Omega$
- At 0°C (ice water)? $6.0\ \Omega$
- At 20°C (room temperature)? $7.0\ \Omega$
- At 100°C (boiling water)? $8.0\ \Omega$

$$R(T) = R_0 [1 + \alpha(T - T_0)]$$

To get α :

- Fit the data (the slope is αR_0). Better method since it uses all the data.
- Use the equation above at a given value of T .

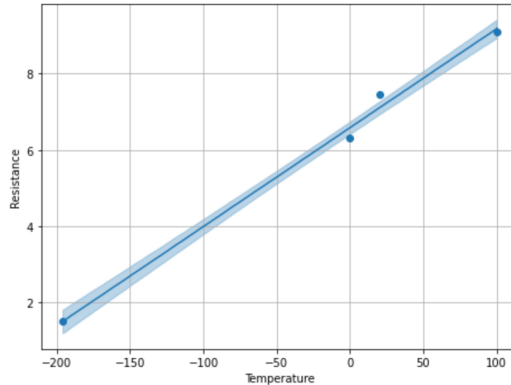
Alpha = 0.0035



The resistivity is linear over 300°C !

Resistance vs Temperature

$\text{Alpha} = 0.0035$



Data from the previous lecture. Resistance is linear from -196°C to 100°C !

Electrical Power

Building Models to Describe our World
Electric Current, Ohm's Law, and Power

Batteries

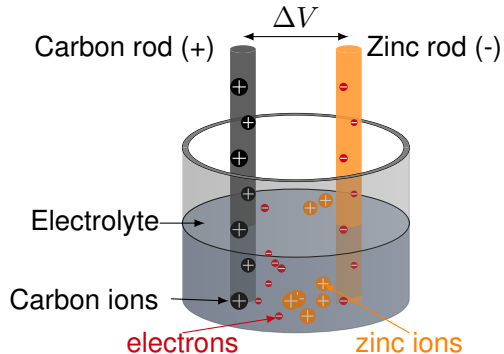
- Batteries have a positive (+) terminal and a negative terminal (−).
- Batteries are characterized by the difference in electric potential between the positive and negative terminal that they provide, e.g. a 9 V battery.
- Most batteries exploit the properties of different materials to create an electrical potential difference.
 - Often, this is done by exploiting the different affinities for materials to be dissolved in acid.



Assorted batteries

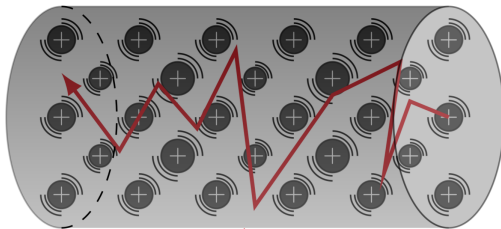
The electric cell

- Zinc “likes” to be dissolved as an ion in acid.
- The zinc ions leave their electrons behind in the electrode, making the zinc electrode negatively charged.
- Electrons that are loosely bound in a carbon electrode are attracted into the electrolyte solution, leaving the carbon electrode positively charged.
- Very quickly, the process stops as an equilibrium is reached, with a net electric potential difference between electrodes.



An electric cell made with an electrolyte, such as H_2SO_4 (sulfuric acid) and two metals.

Power dissipated in a resistor



An electron losing energy in a resistor.

- As electrons move through a resistor, they collide with atoms therein.
- The electric field does work on the electrons, which is then converted into heating up the resistor – no net work is done on the electrons (their average speed is unchanged).
- The rate at which kinetic energy of the electrons is converted into thermal energy is called the “electric(al) power” dissipated in a resistor.

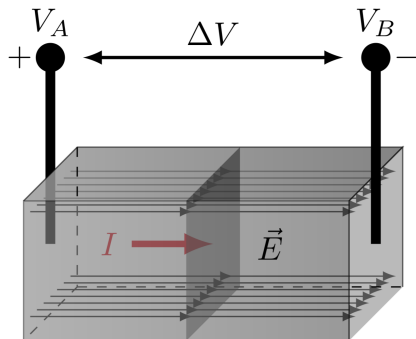
Electric power

- Consider the energy that 1 charge q must deposit in the resistor. The total energy available is the change in potential energy:

$$\Delta U = q\Delta V$$

- If we have many charges going through the resistor, each charge deposits energy E in the form of heat. Then, it makes more sense to define how much *energy per unit time* is deposited in the resistor (which we call power):

$$P = \frac{d}{dt}\Delta U = \frac{d}{dt}q\Delta V = \frac{dq}{dt}\Delta V = I\Delta V$$



A potential difference across a resistor.

Some sloppy notation

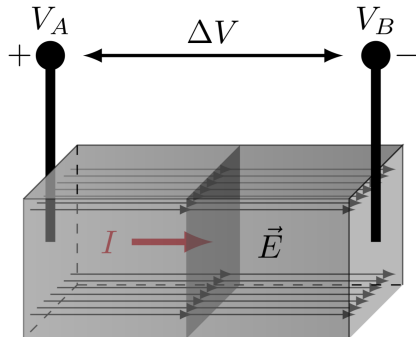
- We apply a potential difference across a resistor, ΔV .
- Ohm's Law tells us how this relates to current and resistance:

$$\Delta V = RI$$

- It is so patently clear that this only makes sense for a potential difference, that all the cool people just drop the Δ and write:

$$V = RI$$

Up to you if you are comfortable with this... The V in the equation above is the *difference* in potential between B and A .



A potential difference across a resistor.

Electric power

- If we have a voltage V , across a resistor R , and current I , then the rate at which heat is dissipated into the resistor is given by:

$$P = IV$$

- Using Ohm's Law to get rid of I or V , we can rewrite this as:

$$P = I^2 R \quad (\text{say, we know the current through a resistor})$$

$$P = \frac{V^2}{R} \quad (\text{say, we know the voltage across a resistor})$$

Household appliances

- Household appliances are designed such that they will work with a fixed voltage difference (120 V in North America).
 - Often appliances have specifications about how much power they use:
 - 1500 W microwave.
 - 60 W lightbulb.
 - 1200 W hair dryer.
- These specifications are always assuming that they have a constant voltage applied to them (120 V in North America).



https://commons.wikimedia.org/wiki/File:Gluehlampe_01_KMJ.ica



<https://www.ebay.ca/itm/333729494226>

Alternating current/voltage - Why?!

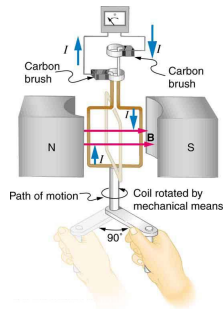
- Electricity (electric power) is useful, right?
- So we need efficient ways to produce and transmit it!
- We can generate DC current (giant batteries), but it's not efficient or practical for the amount of energy that our society uses.
- It's very easy to generate AC current (or AC voltage) using a generator.

Electric generator

- In a few weeks, you will learn about electromagnetic induction:
 - If we change the magnetic flux through a coil, we can generate a voltage.
- If we rotate a coil in a fixed magnetic field, we will generate a sinusoidal voltage:

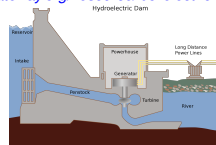
$$V = V_0 \sin(\omega t)$$

- Easy to build turbines (things that spin when we run water or hot gases through them)
 - Convert mechanical or thermal energy into electric potential energy



[https:](https://www.texasgateway.org/resource/65-electric-generators)

[//www.texasgateway.org/resource/65-electric-generators](https://www.texasgateway.org/resource/65-electric-generators)



https://energyeducation.ca/encyclopedia/Hydroelectric_facility

AC Voltage and current

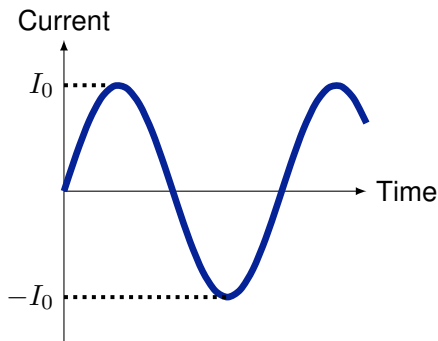
- Ohm's law still true, so across a resistor with sinusoidal voltage:

$$V = V_0 \sin(\omega t)$$

$$V = RI$$

$$I = \frac{V}{R} = \frac{V_0}{R} \sin(\omega t) = I_0 \sin(\omega t)$$

- The average current/voltage is 0.



Power dissipated in a resistor with AC current

- If the current is given by:

$$I(t) = I_0 \sin(\omega t)$$

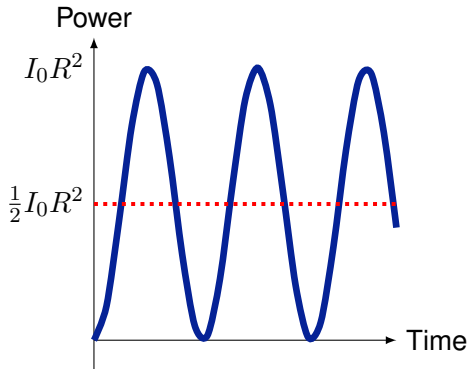
- the instantaneous power dissipated in the resistor (at some time, t) is:

$$P(t) = I^2(t)R = I_0^2 \sin^2(\omega t)R$$

- The average power is non-zero:

$$\bar{P} = I_0^2 R \overline{\sin^2(\omega t)} = \frac{1}{2} I_0^2 R$$

(The average value of $\sin^2(\omega t)$ over one period is $\frac{1}{2}$.)



With alternating current, the instantaneous dissipated power is always greater than zero (makes sense).

RMS Current

- For sinusoidal current, with maximum amplitude I_0 , we can write the average dissipated power as:

$$\bar{P} = \frac{1}{2} I_0^2 R$$

- Would like to have some sort of “effective” current, I_{eff} , so that we can use the “usual” formula for power:

$$\bar{P} = I_{eff}^2 R$$

- This works if we introduce an “effective current”, I_{eff} , called the RMS current:

$$I_{eff}^2 = \frac{1}{2} I_0^2$$

$$\therefore I_{eff} = \frac{1}{\sqrt{2}} I_0 = I_{rms}$$

I_{rms} is the average value of the square of the current, the *square root of the mean of the squared current*: $I_{rms}^2 = I_0^2 \overline{\sin^2(\omega t)} = \frac{1}{2} I_0^2$

RMS Current and voltage

- We can define *RMS values* of the current and voltage:

$$I_{rms} = \frac{1}{\sqrt{2}} I_0 \quad V_{rms} = \frac{1}{\sqrt{2}} V_0$$

- This allows us to easily calculate the average power dissipated by a resistor when it has AC voltage (current) across (through) it, using the RMS values:

$$V_{RMS} = RI_{rms} \quad \bar{P} = I_{rms}^2 R \quad \bar{P} = I_{rms} V_{rms} \quad \bar{P} = \frac{V_{rms}^2}{R}$$

- Remember, the *instantaneous power* changes as a function of time, so it's more useful to think about the average power that our appliances use (e.g. how much will it cost to have a 100 W light bulb on all the time).